Ethical and legal declarations

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Manuscript: Disentangling spatial-filter and classifier contributions to limited-label motor-imagery EEG recalibration

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Research ethics

The present secondary analysis was exempt from additional ethical review under Article 32, item (1), of China's Measures for Ethical Review of Life Science and Medical Research Involving Human Participants (2023). It used lawfully obtained publicly available data, caused no harm to participants, and did not involve sensitive personal information or commercial interests. The original Kumar data collection reported University of Texas at Austin approval under protocol 2020-03-0073. The reviewed BNCI2014-001 source publication and dataset description did not supply a dataset-2a-specific committee, approval number or consent statement; their data-collection ethics details are not inferred from another dataset. Ziyuan Zhu had no access to identifying participant information and made no attempt to re-identify participants.

Informed consent

The original Kumar study reported written informed consent before data collection. No dataset-2a-specific consent statement was identified in the reviewed BNCI source documents. The present study analyzed coded public records; no new participants were recruited or contacted and no additional consent was collected.

Author contributions

All authors have accepted responsibility for the entire content of this manuscript and approved its submission.

Ziyuan Zhu: Conceptualization; Methodology; Software; Formal analysis; Validation; Visualization; Writing – original draft; Writing – review & editing.

Sihao Liu assisted with code verification, organization of project files, and checking the journal's submission policies.

Haoyang Yu assisted with data checking and aggregation, project-file organization, and GitHub uploads.

Use of large language models, AI and machine learning tools

OpenAI Codex assisted with code revision, GitHub updates, file compilation and organization, manuscript preparation and saved-result checks; OpenAI ChatGPT assisted with grammatical correction. The authors are responsible for verifying sources, scientific claims and the final manuscript.

Conflict of interest

The authors state no conflict of interest.

Research funding

None declared.

Data availability

The source dataset is publicly available as Kumar2024, NEMAR nm000177 version 1.0.5, https://doi.org/10.82901/nemar.nm000177.v1.0.5. BNCI2014-001 original MAT recordings are available through https://bnci-horizon-2020.eu/database/data-sets/001-2014/ under CC BY-ND 4.0; frozen NEMAR nm000139/v1.0.3 provenance supplies their exact byte hashes. This revision adds external trial identities, all four methods' saved predictions, participant/run tables and fixed-protocol code. Original component-analysis predictions and metadata remain in the previously versioned submission data package. The V3 saved predictions, metrics, text training histories and diagnostics are publicly archived at https://github.com/jackzhu119/few-shot-mi-eeg/tree/500831c086d9e1d1e8412cf8ebca5fb6a288db76/training_status/paper2-discovery-v3/result_exports/83cbdfdcfa541b3ab3addfc3 . This revision supplies the complete participant-level comparisons, figure datasets and saved-result reconstruction code. Original data rights and third-party licenses remain applicable.

Software availability

Original analysis code is available at https://github.com/jackzhu119/few-shot-mi-eeg under the MIT license. Original derived data, figures and manuscript content use CC BY 4.0. The adapted citation style retains its CC BY-SA 3.0 license.